



Analisis Kinerja Arsitektur Integrasi Data ETL, ELT, dan Hibrida pada *Data Warehouse* Berbasis *PostgreSQL*

Luthfan Aryananda Purwito (5026221166)

Dosen Penguji 1:

Radityo Prasetianto Wibowo, S.Kom., M.Kom.

Dosen Penguji 2:

Irmasari Hafidz, S.Kom., M.Sc.

Dosen Pembimbing:

Faizal Johan Atletiko, S.Kom., M.T.



Apa saja yang akan dibahas?

- BAB 1** Pendahuluan
- BAB 2** Tinjauan Pustaka
- BAB 3** Metodologi



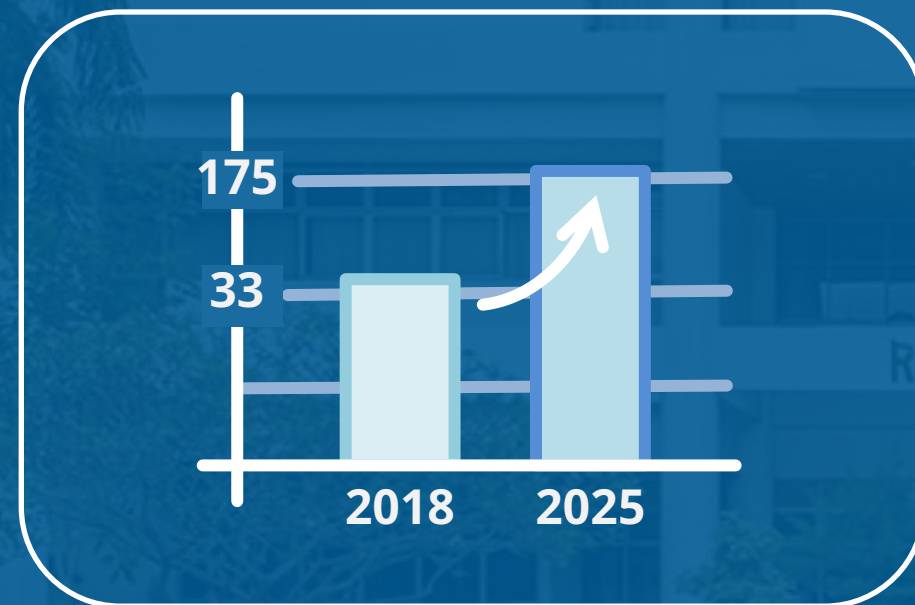


BAB 1

Pendahuluan

REKTORAT

Apa yang melatarbelakangi tugas akhir ini?



Lonjakan masif pada **volume** dan **varietas** data (terstruktur, semi-terstruktur, dan tidak terstruktur)



Organisasi dituntut memiliki strategi integrasi data yang **efisien** dan **cepat** untuk pengambilan keputusan

Apa yang melatarbelakangi tugas akhir ini?



ETL menghadapi **isu skalabilitas** dan **kinerja** saat menangani data bervolume besar



ELT menawarkan **performa lebih tinggi** dengan memanfaatkan kekuatan komputasi warehouse.



Hibrida menggabungkan **kontrol ETL** dan **kecepatan ELT**.

Apa yang melatarbelakangi tugas akhir ini?



Studi komparatif komprehensif
antara ketiga arsitektur ini
masih terbatas...



...terutama implementasi
skenario *on-premise* dengan
variasi format data
(CSV, JSON, Parquet)

Rumusan Masalah

- 01 Bagaimana perbandingan **waktu pemrosesan total** antara arsitektur ETL, ELT, dan hibrida dalam menangani beban kerja dengan bentuk data terstruktur dan semi-terstruktur?
- 02 Bagaimana perbandingan **efisiensi penggunaan sumber daya (CPU dan memori)** pada ketiga arsitektur tersebut selama proses integrasi data berlangsung?
- 03 Bagaimana perbandingan **efisiensi penggunaan ruang penyimpanan disk** pada ketiga arsitektur tersebut pada tahap penyimpanan data mentah dan setelah dimuat ke data warehouse?

Tujuan

- 01 Untuk membandingkan **waktu pemrosesan total** antara arsitektur ETL, ELT, dan hibrida dalam menangani beban kerja dengan bentuk data terstruktur dan semi-terstruktur
- 02 Untuk membandingkan **efisiensi penggunaan sumber daya (CPU dan memori)** pada ketiga arsitektur tersebut selama proses integrasi data berlangsung
- 03 Untuk membandingkan **efisiensi penggunaan ruang penyimpanan disk** pada ketiga arsitektur tersebut pada tahap penyimpanan data mentah dan setelah dimuat ke data warehouse

Batasan Masalah



- 01 **Dataset TPC-H** yang dikonversi menjadi dua jenis:
 - Terstruktur (CSV, Parquet)
 - Semi-Terstruktur (JSON)
- 02 **Metrik Pengujian:**
 - Waktu pemrosesan total
 - Penggunaan CPU
 - Penggunaan memori
 - Penggunaan ruang penyimpanan
- 03 **Platform dan teknologi:**
 - Data Warehouse: PostgreSQL
 - Arsitektur ETL: Python
 - Arsitektur ELT: Python + dbt
 - Arsitektur Hibrida:
 - Extract, Clean, dan Load dengan Python
 - Transform dengan dbt

Manfaat

01

Praktis

Memberikan **rekomendasi teknis** berbasis data dalam memilih arsitektur integrasi data yang paling optimal, berdasarkan karakteristik data dan kebutuhan spesifik mereka.

02

Akademis

Menyajikan **wawasan dan data perbandingan kuantitatif** antara arsitektur ETL, ELT, dan hibrida yang dapat mengisi kesenjangan riset di bidang integrasi data berskala besar.



BAB 2

Tinjauan Pustaka

REKTORAT



Penelitian Terdahulu



A Model for Enhancing Unstructured Big Data Warehouse Execution Time

Marwa Salah Farhan, Amira Youssef, Laila Abdelhamid

- Perbandingan kinerja kuantitatif antara ETL, ELT, dan beberapa jenis hibrida.
 - Dilakukan secara *on-premise*
 - Metrik yang digunakan:
 - Waktu eksekusi
 - Penggunaan memori
- (2024)



ETL vs. ELT: Optimizing Data Integration for Retail and Insurance Analytics

Venkatesha Prabhu Rambabu, Chandrashekar Althathi, Amsa Selvaraj

- Perbandingan ETL dan ELT
 - Metrik yang digunakan:
 - Kinerja
 - Skalabilitas
 - Efisiensi sumber daya
- (2023)



Penelitian Terdahulu

2020 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS)

Comparison of the E-LT vs ETL Method in Data Warehouse Implementation: A Qualitative Study

1st Edward Manopo Haryono
Faculty of Computer Science
Universitas Indonesia
Jakarta, Indonesia
edward.manopo@ui.ac.id

2nd Edna
Faculty of Computer Science
Universitas Indonesia
Jakarta, Indonesia
edna92@ui.ac.id

3rd Adi Setiawan Tri W
Faculty of Computer Science
Universitas Indonesia
Jakarta, Indonesia
adi.setiawan72@ui.ac.id

4th Indira Gusman
Faculty of Computer Science
Universitas Indonesia
Jakarta, Indonesia
indira.gusman@ui.ac.id

5th Ashadul Nisar Hidayat
Faculty of Computer Science
Universitas Indonesia
Jakarta, Indonesia
ashadul@ui.ac.id

6th Untung Rahajaja
Universitas Rahajaja
Tangerang, Indonesia
untung@rahajaja.info

Abstract — The data warehouse is an enterprise system that archives various kinds of data from various main systems and integrates them into a place. There are two data processing methods in implementing a data warehouse, namely ETL (Extract, Transform, Load) and E-LT (Extract, Load, Transform). In determining the implementation of a data warehouse, it is often difficult to choose the appropriate method. Because this is related to Cost, Process, Performance, and Continuous Improvement in a company. This paper discusses the implementation of a data warehouse in various industries by conducting a survey to compare the use of ETL and E-LT methods. The survey was conducted on people with backgrounds in work related to data warehouse from various industries. Based on the results of this survey, it is hoped that a better method of implementing a data warehouse can be obtained.

Keywords — data warehouse, ETL, E-LT

I. INTRODUCTION

In modern times, many companies need a data warehouse. The data warehouse is a collection of data that is subject-oriented, integrated, and not easily changed in support of the management decision-making process [1]. Data Warehouse is not only data storage but also for the needs of accurate data analysis for the future progress of the company. The right and very competitive business competition requires every company to be able to look for good opportunities, especially in making the right decisions so that they can continue to improve the company's performance in the future. Companies can improve the quality and make business decisions based on data-based systems, which means it is also a form of activity or implementation which is used to support the progress of the company to process data and information which is then processed again into knowledge so that later it can be used for the analysis process for the company more broadly again. Data is also useful for companies to be able to predict how the company's sales will be in a certain period and how the company can increase its profits/profits in the future. The company data warehouse provides strategic information to support decision making [2]. Making the right decision must be supported by good data analysis. Data is a valuable asset for companies that can provide benefits in

operational and strategic aspects of organizational business development [3]. Good data quality is the most important factor in the success of a data warehouse. Integrated data in the data warehouse allows companies to present information quickly, accurately, and in real-time to support decision-making processes and company policies. Besides, to their competitors and to anticipate trends in business competition between companies engaged in the same business field.

In the data warehouse itself, there are two data integration methods, namely ETL (Extract, Transform, Load) and E-LT (Extract, Load, Transform). This integration method plays a very important role in building a data warehouse. This is because the data that is pulled into the data warehouse comes from various heterogeneous sources. In several previous studies, the ETL Process has also been discussed as part of the concepts and basics of data warehouse and OLAP [4], a data warehouse model that uses the ETL process [5], conducts a comparative study between ETL and E-LT to load data into data warehouse [6], to conduct a comparative study between ETL and E-LT in a data warehouse using various parameters [10]. From these studies, there is no research related to the comparison of the ETL and E-LT methods at the time of implementing the data warehouse using a qualitative approach. Therefore, the focus of this research itself is to compare the ETL and E-LT methods as data integration methods when implementing a data warehouse using qualitative methods, so professionals working in the data warehouse field with different industrial backgrounds.

Furthermore, the structure of this writing is structured as follows: the next section is a study of relevant literature in the process. The third part, research methodology and provides data analysis. The fourth part, results, and discussion. The fifth part, the conclusions obtained from this study.

II. STUDY OF LITERATURE

This section contains a literature review on data warehouses and more specifically about data processing methods in implementing a data warehouse. The data warehouse is a data repository that comes from the integration of various operational data sources managed in various business units of the organization [4], which aims to

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Comparison of the E-LT vs ETL Method in Data Warehouse Implementation: A Qualitative Study

Haryono et al.

- Perbandingan kualitatif antara ETL dan ELT
- Metrik yang digunakan: Kinerja, dukungan data tidak terstruktur, kompleksitas, dan biaya

(2020)

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Optimizing ETL Processes for High-Volume Data Warehousing in Financial Applications

Samyukta Rongala
Alumnus of University of Missouri Saint Louis
Email: rsamyukta@gmail.com

ARTICLE INFO

ABSTRACT

Keywords: ETL, optimization, financial data warehousing, high-volume processing, Big Data, cloud computing, compliance, performance metrics.

1. INTRODUCTION

A financial company deals with a vast and varied amount of data, including transactions, market feeds, regulatory reports, and customer interactions. ETL (Extract, Transform, Load) processes play a crucial role in extracting data from multiple sources, cleaning it, and converting it into a structured format that can be loaded into data systems for further use (Raka, 2024). Given that financial data is highly sensitive, and the industry demands both high performance and strict adherence to regulatory standards, the accuracy and speed of ETL processes are critical. The sector relies on fast and accurate data processing for real-time decision-making, continuous monitoring of operations, detecting fraud and illegal activities, and meeting legal requirements. As financial organizations face growing volumes of data, managing this data efficiently through ETL processes becomes even more important (Noshaki, 2023). This research focuses on the challenges of high-velocity financial data processing and examines issues such as scalability, performance, and compliance with regulations. It also seeks to identify opportunities for improvement and propose innovative solutions (Salazar, 2026). By studying technological advancements and making recommendations, the research aims to bridge the gap between current ETL practices and the future needs of financial data management. Additionally, it emphasizes the importance of standardizing ETL processes to improve data usability and efficiency, helping financial institutions stay competitive in an increasingly data-driven world.

2. LITERATURE REVIEW

2.1 Overview of ETL Processes and Architectures

Comparison of ETL Architectures

Figure 1: ETL Architectures

In figure 1 the comparison of ETL architectures highlights the differences in latency and throughput between batch processing, real-time processing, and ETL pipelines. ETL (Extract, Transform, Load) operation is considered as fundamental to data warehouse system as it provides both the mechanism for obtaining data as well as the means of

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Optimizing ETL Processes for High-Volume Data Warehousing in Financial Applications

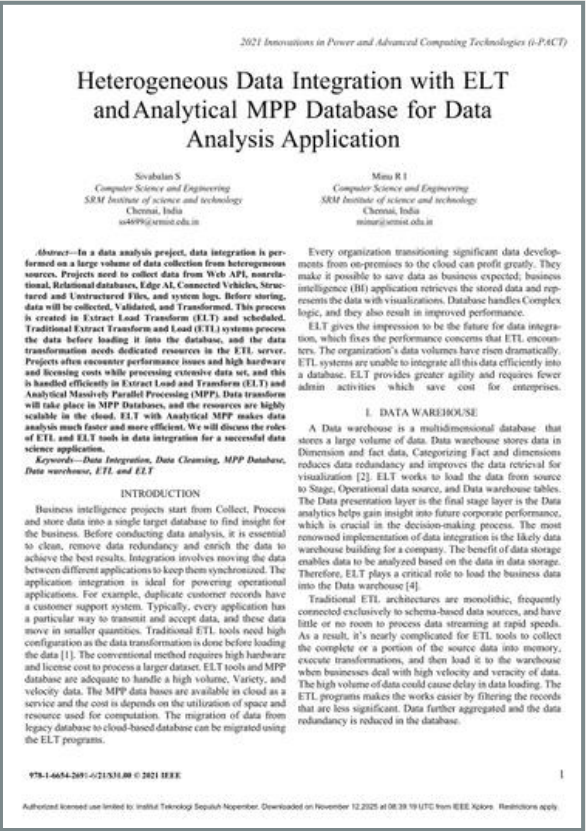
Samyukta Rongala

- Evaluasi dan optimalisasi kinerja ETL
- Metrik yang digunakan:
 - Kinerja secara kuantitatif
- Menyajikan bukti visual kinerja ETL

(2025)



Penelitian Terdahulu



Heterogeneous Data Integration with ELT and Analytical MPP Database for Data Analysis Application

Sivabalan S., Minu R. I.

- Perbandingan kinerja ETL dan ELT
- Mengidentifikasi biaya perangkat
- Mengidentifikasi penyebab bottleneck ETL

(2021)



Perbandingan Performa Teknologi Data Lake dan Data Warehouse dalam Pengelolaan Data Stack Overflow Menggunakan Arsitektur Hybrid Hub-and-Spoke dan Kimball Method

Ihsan Kamil Al Ghozi

- Menggunakan ETL untuk integrasi data
- Metrik yang digunakan: Durasi proses ETL, Efisiensi sumber daya
- Dilakukan secara lokal (on-premise)

(2025)



Dasar Teori

5V

1. **Volume** : Jumlah data yang sangat besar
2. **Variety** : Keragaman tipe dan sumber data, mulai dari data terstruktur hingga data tidak terstruktur
3. **Velocity** : Kecepatan akuisisi dan pemrosesan data
4. **Veracity** : Kualitas, keakuratan, dan keandalan data
5. **Value** : Potensi nilai atau wawasan dan pengetahuan yang bisa diambil untuk pengambilan keputusan

Data Lake

Repository yang dapat menyimpan **data mentah** dengan **volume yang besar**, dan dibantu oleh deskripsi metadata yang lengkap.

Data yang dapat disimpan:

- Terstruktur
- Semi-terstruktur
- Tidak terstruktur

Risiko:

Dapat menjadi *Data Swamp* jika tidak dikelola dengan baik.

Data Warehouse

Repository yang dirancang untuk menyimpan data yang bersifat **nonvolatil, terintegrasi, bervariasi secara waktu, dan berorientasi subjek**. (Harby & Zulkernine, 2025)

Data warehouse mengikuti prinsip **OLAP** (*Online Analytical Processing*)

Tabel fakta

Tabel dimensi





Dasar Teori

Extract, Transform, Load (ETL)

Proses integrasi data yang memindahkan dan menyiapkan data untuk kebutuhan analisis

Extract

Memasukkan data ke sistem ETL, untuk kebutuhan manipulasi data.

Transform

Biasanya mencakup pembersihan, penggabungan, agregasi, dan sejenisnya.

Load

Pemuatan data ke dalam model dimensional di *data warehouse*.

Extract, Load, Transform (ELT)

Proses integrasi data di mana data mentah langsung dimuat ke sistem target

Extract

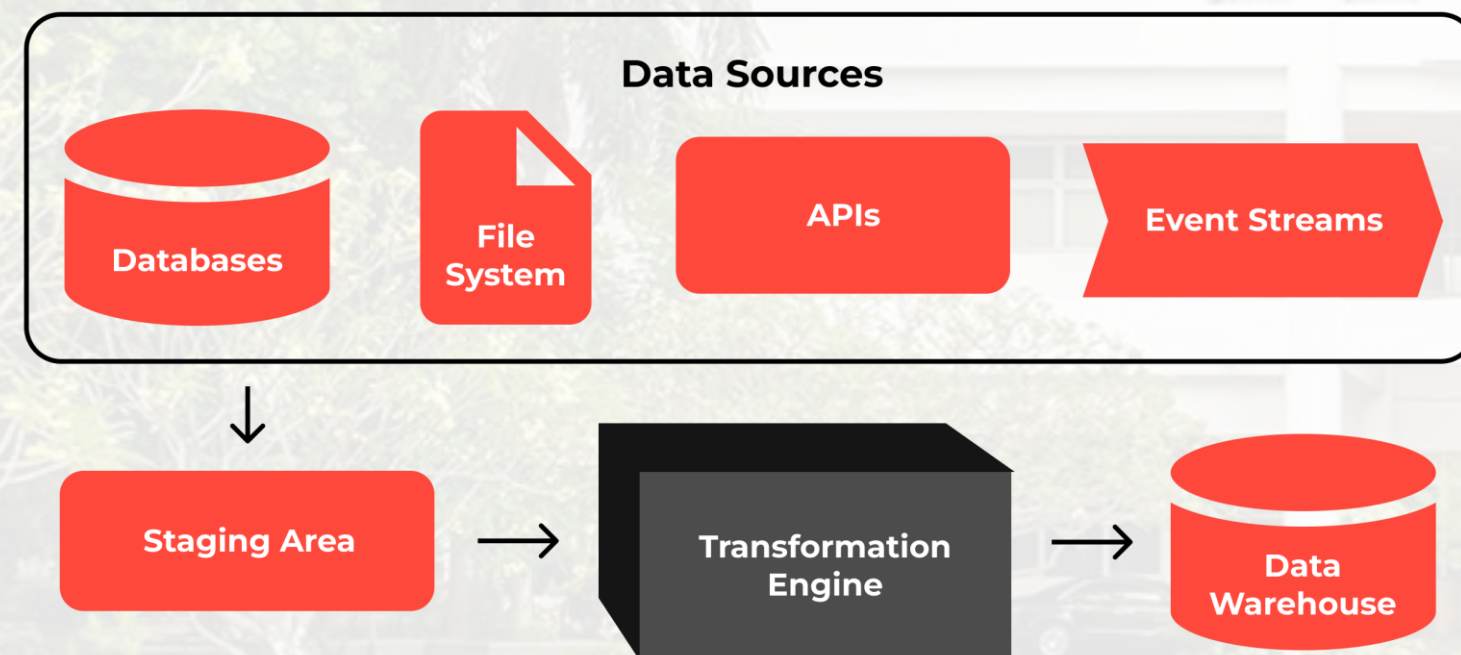
Mengekstraksi data dari sistem sumber aslinya secara efisien.

Load

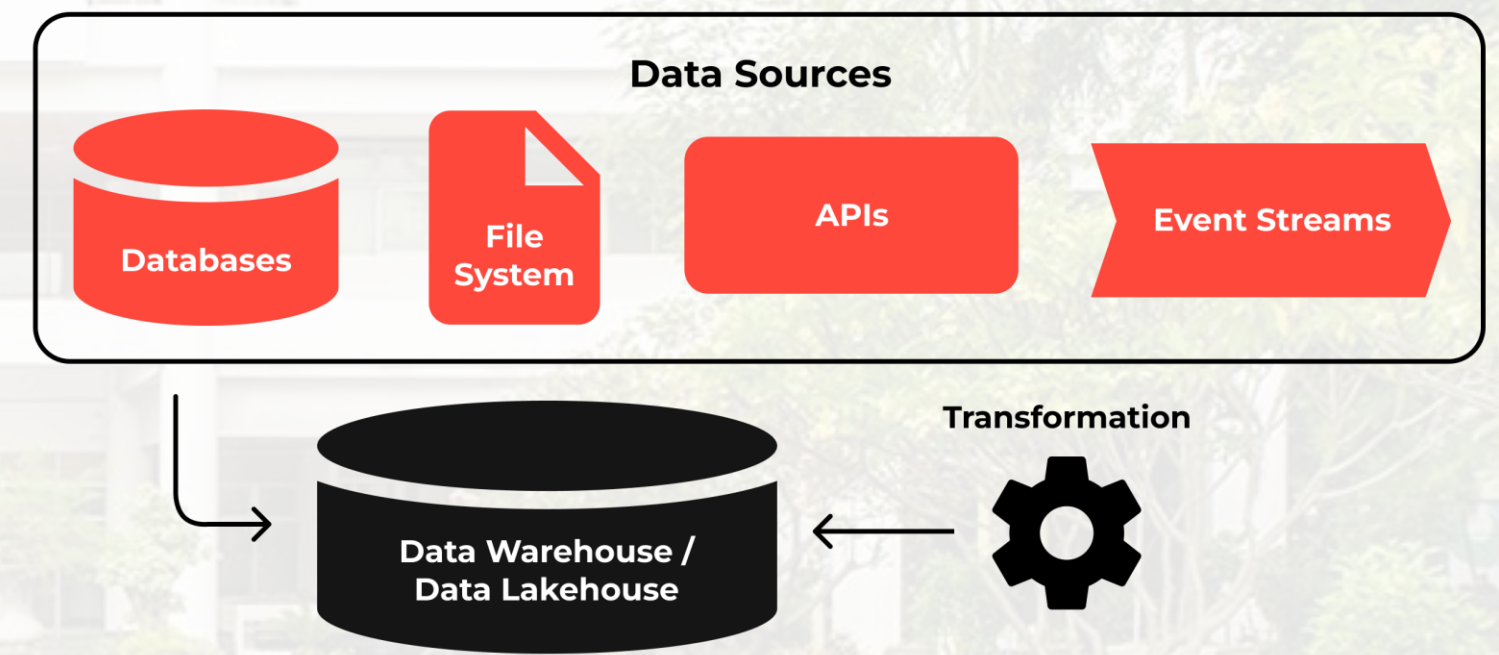
Pemuatan data mentah langsung ke *data warehouse* tujuan.

Transform

Pemfilteran, penggabungan, agregasi, dan transformasi analitis lainnya.



(DataForge, 2023)



(DataForge, 2023)

Dasar Teori

Arsitektur Hibrida

Proses integrasi data yang memanfaatkan kekuatan kedua model ETL & ELT.

ETL

- Tata kelola
- Kepatuhan
- Kemampuan untuk transformasi data secara kompleks

ELT

- Semi-terstruktur dan tidak terstruktur
- Bervolume besar
- Mampu menangani real-time

ECLT (Farhan et al., 2024):

Menerapkan **transformasi ringan** (seperti pembersihan data) **sebelum data dimuat** ke dalam sistem target

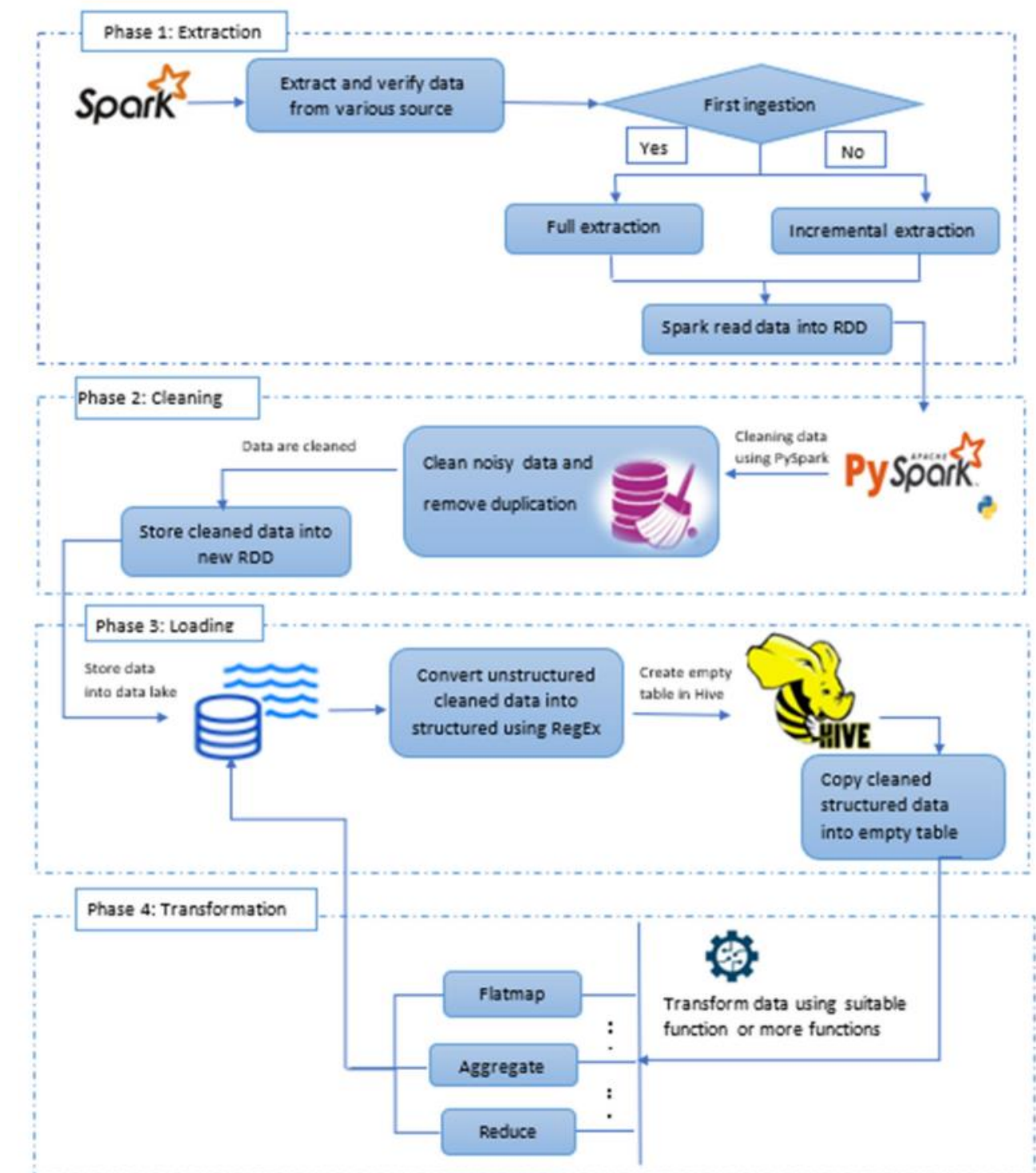
Memanfaatkan kekuatan *engine* sistem tujuan untuk **transformasi yang lebih berat** (seperti agregasi, penggabungan)

Extract

Clean

Load

Transform



(Farhan et al., 2024)



Dasar Teori

Alat Integrasi Data

PostgreSQL

dbt

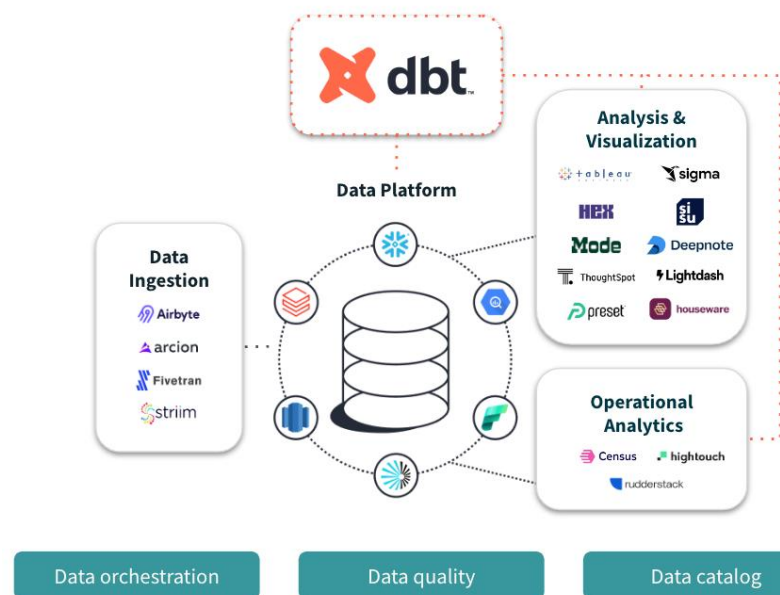
Python

Format Penyimpanan Data

CSV

JSON

Apache
Parquet



(dbt Labs, n.d.)

dbt (Data Build Tool)

Alat *open-source* modern yang secara khusus dirancang untuk membantu analis data melakukan transformasi data, terutama dalam penerapan metode ELT (extract, load, transform).

Apache Parquet

Format penyimpanan **kolumnar** yang dirancang khusus untuk menyimpan, mengelola, dan menganalisis data bervolume besar secara efisien.

Single Parquet File

Row Group
*size = 2 rows

Row Group
*size = 1 rows

Column

Column

Column

Column

Column

Column

a

b

2020-01

2020-02

1

2

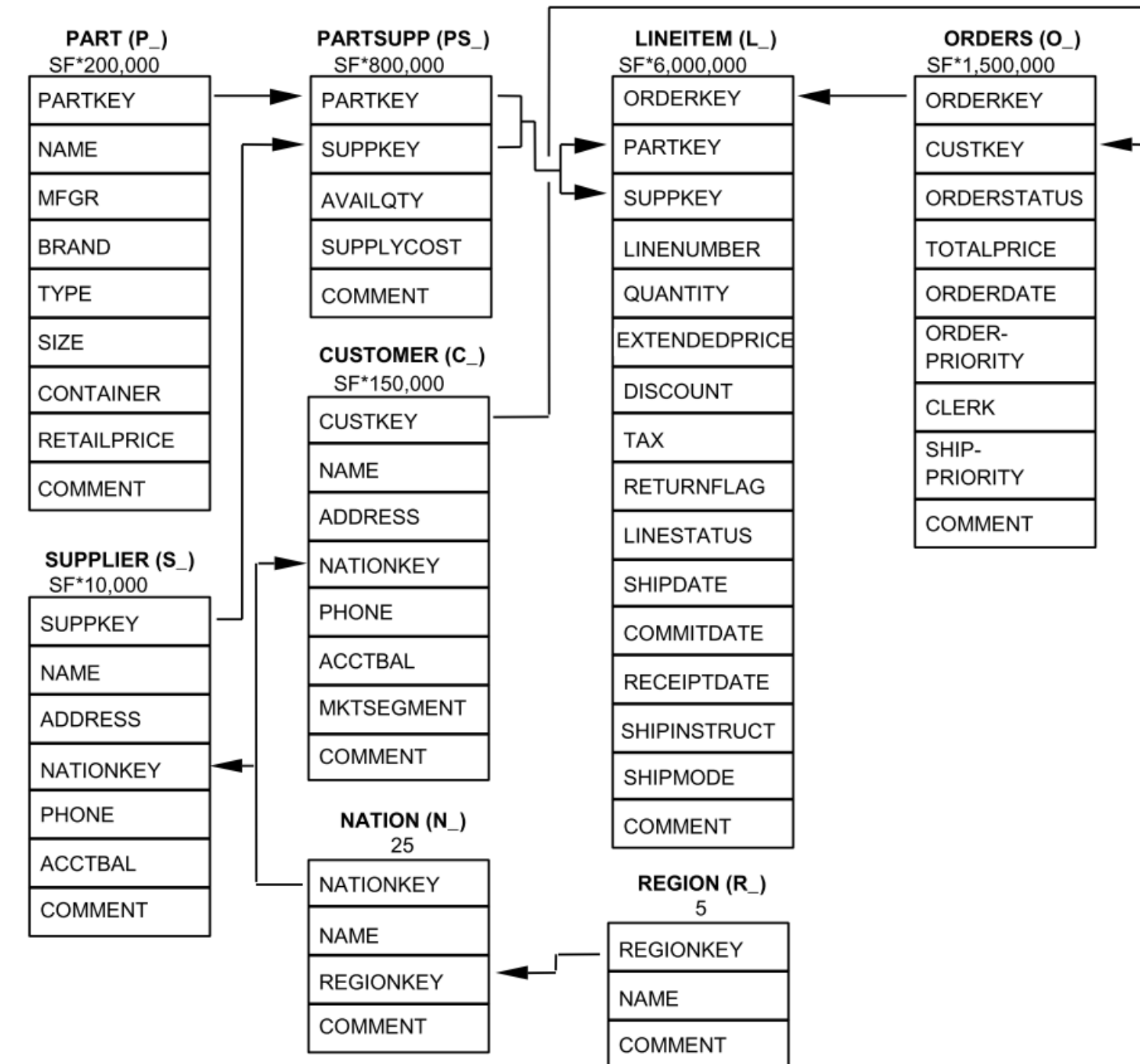
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3

2020-03

TPC-H (TPC Benchmark™ H)

- Merupakan salah satu **benchmark standar industri** yang dikembangkan oleh Transaction Processing Performance Council (TPC).
- Dirancang khusus **untuk mengukur kinerja sistem basis data** terutama dalam menangani tantangan di lingkungan sistem pendukung keputusan modern.
- TPC memiliki **program generator data resmi** yang dinamakan sebagai **DBGEN**.
- Pembuatan volume data dikontrol dengan parameter yang disebut dengan **Scale Factor (SF)**.





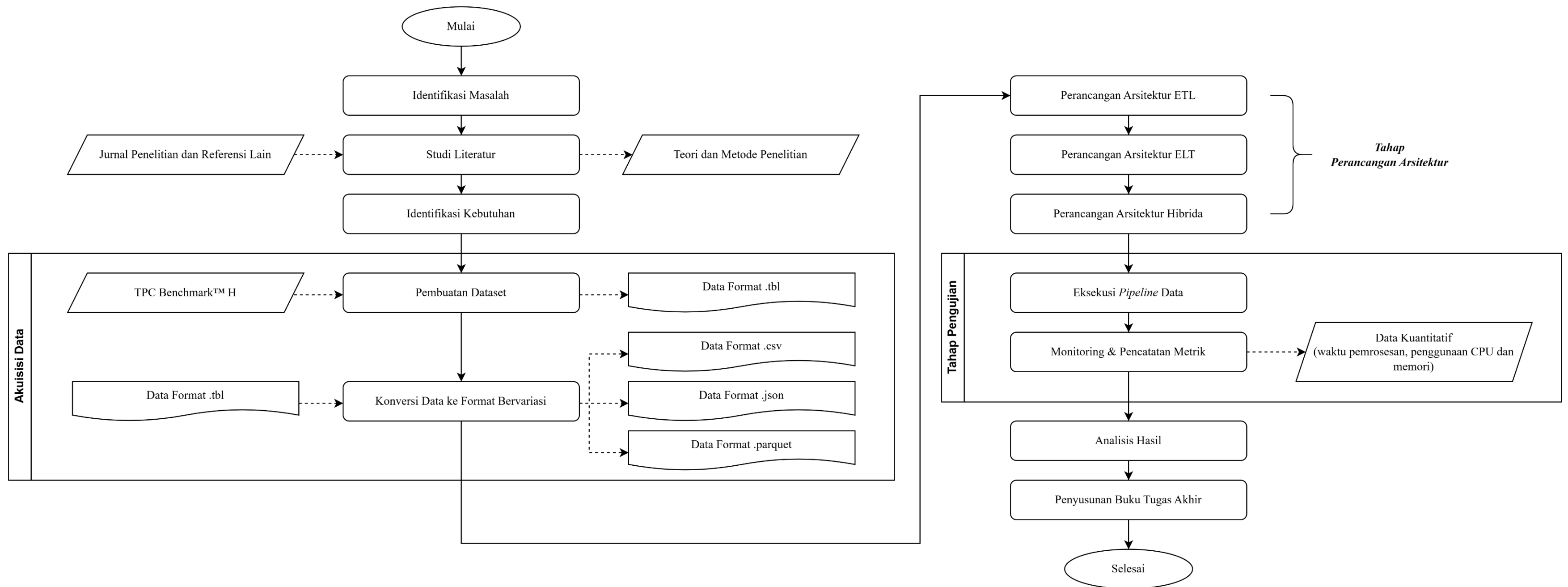
BAB 3

Metodologi

REKTORAT



Metodologi Pelaksanaan

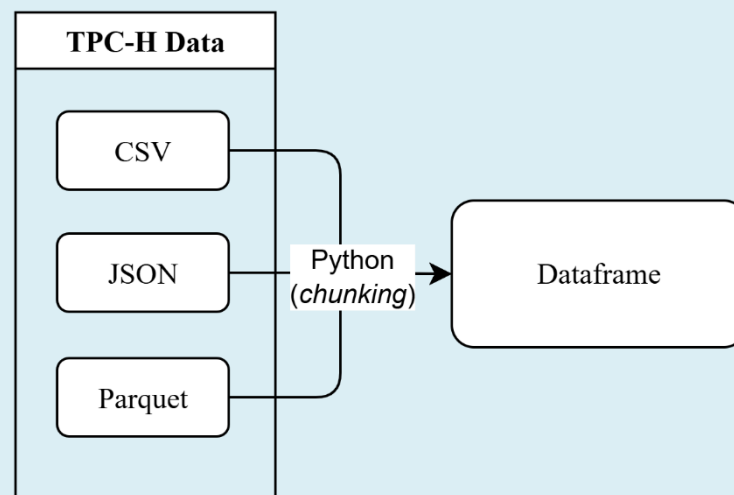


Arsitektur ETL

1

Extract

- Membaca data sumber TPC-H yang sudah dikonversi menjadi format CSV, JSON, dan Parquet
- Ekstraksi dilakukan menggunakan Python secara bertahap (*chunking*)



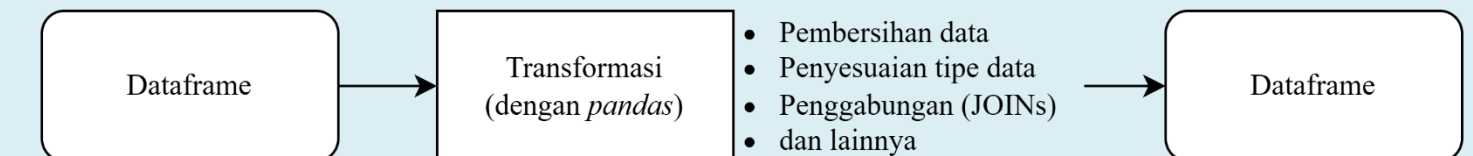
Server

2

Transform

Melakukan seluruh transformasi, seperti:

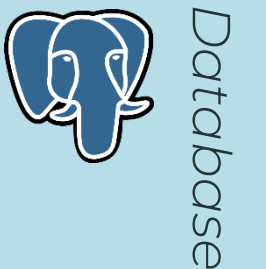
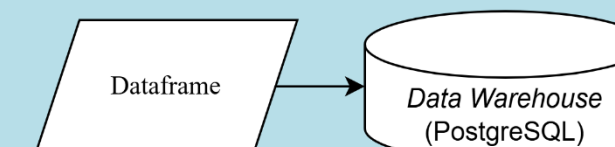
- Pembersihan data (penanganan duplikat, *null*)
- Penyesuaian tipe data
- Penggabungan (*JOIN*)
- Agregasi
- Pemfilteran



3

Load

Tahap **pemuatan data** ke dalam sistem basis data PostgreSQL target

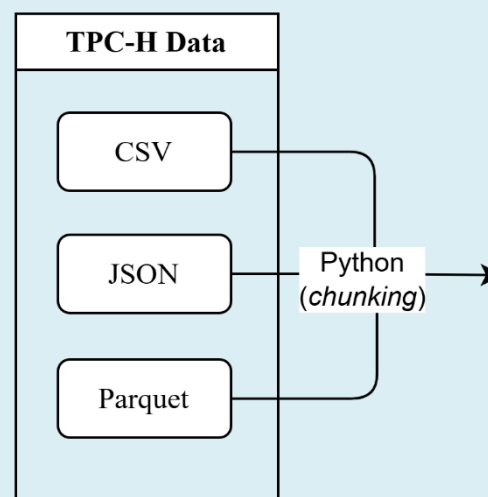


Arsitektur ELT

1

Extract

- Membaca data sumber TPC-H yang sudah dikonversi menjadi format CSV, JSON, dan Parquet
- Ekstraksi dilakukan menggunakan Python secara bertahap (*chunking*)



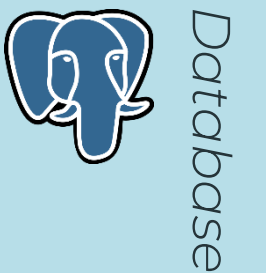
Server

2

Load

Tahap **pemuatan data** ke dalam sistem basis data PostgreSQL target.

Masih berupa data mentah



3

Transform

Melakukan **seluruh transformasi**, seperti:

- Penggabungan (*JOIN*)
- Agregasi
- Pemfilteran

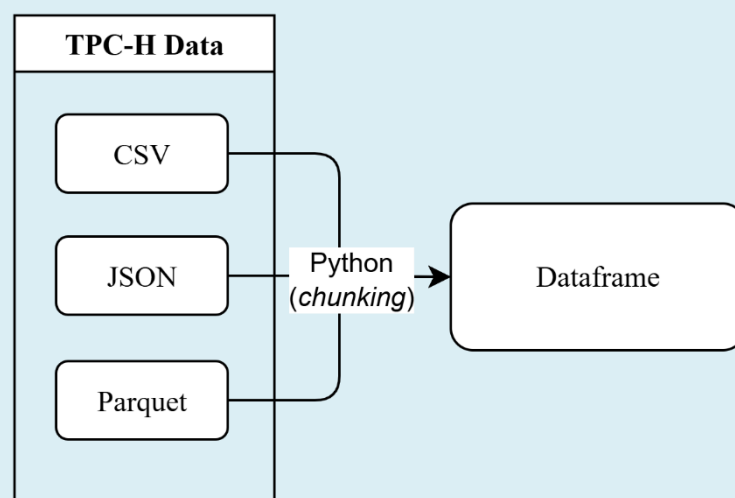


Arsitektur Hibrida

1

Extract

- Membaca data sumber TPC-H yang sudah dikonversi menjadi format CSV, JSON, dan Parquet
- Ekstraksi dilakukan menggunakan Python secara bertahap (*chunking*)



Server

2

Clean

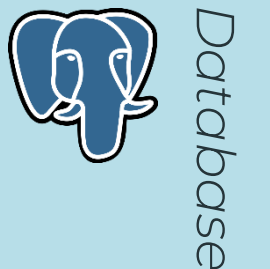
Tahap pembersihan data dan transformasi ringan

- Standarisasi nama kolom
- Penyesuaian tipe data
- Penghapusan duplikat

3

Load

Tahap **pemuatan data** ke dalam sistem basis data PostgreSQL target.



4

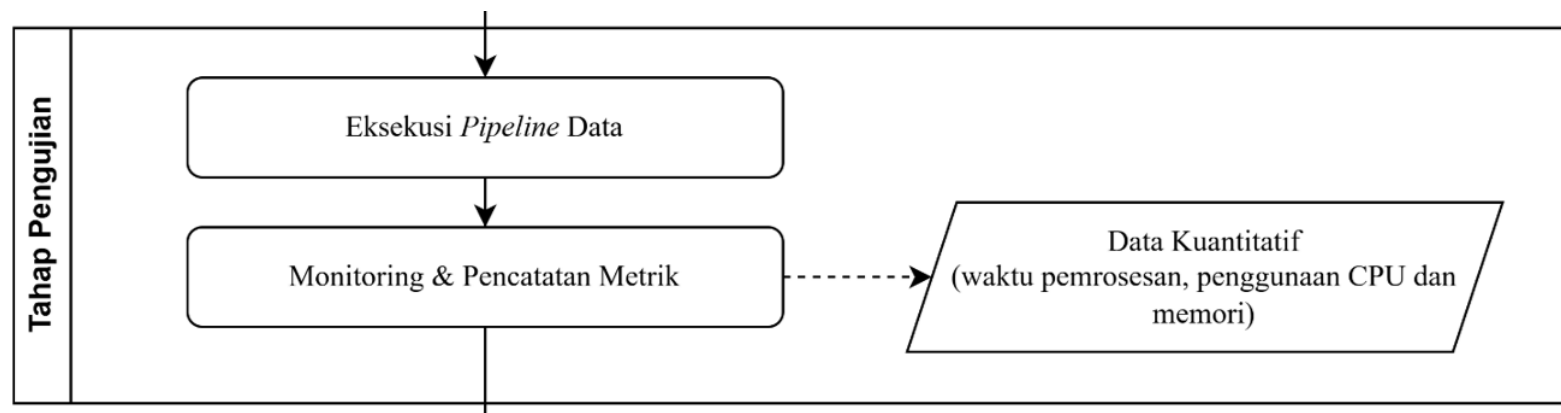
Transform

Melakukan **transformasi berat**, seperti:

- Penggabungan (*JOIN*)
- Agregasi
- Pemfilteran



Tahap Pengujian



Eksekusi *Pipeline*

Arsitektur	Format Data
Extract, Transform, Load (ETL)	CSV
Extract, Load, Transform (ELT)	JSON
Hibrida	Parquet

Monitoring & Pencatatan Hasil

- **Waktu** Pemrosesan (detik)
- Pengujian **Efisiensi Sumber Daya**



No	Jenis Kegiatan	Bulan / Minggu																			
		Bulan ke-1				Bulan ke-2				Bulan ke-3				Bulan ke-4				Bulan ke-5			
		I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV
1	Studi Literatur																				
2	Identifikasi Kebutuhan																				
3	Pengumpulan Data																				
4	Perancangan Arsitektur ETL																				
5	Perancangan Arsitektur ELT																				
6	Perancangan Arsitektur Hibrida																				
7	Tahap Pengujian																				
8	Tahap Analisis Hasil																				
9	Penyusunan Tugas Akhir																				



Sekian,
Terima Kasih

REKTORAT